

Departure From the Photon Baseline:
A Universal Ruler for Mass, Gravity, and Time

The 27 Lagrangians of Gravity

A Survey of the Field's Standards of Evaluation

Dervis Kadunic

Paper 1 of the Series

Abstract. When a new gravity framework is proposed, by what standard should it be evaluated? This paper surveys twenty-seven published Lagrangian approaches to gravity, from Einstein-Hilbert to Horndeski to Loop Quantum Gravity, and catalogs what each framework adopts as input — a functional form, a symmetry, a coupling constant, a discretization choice — versus what it derives or fits to observation. The honest observation is that every gravity framework in the field has the same structure: adopt a form, derive consequences, match data. This is not a criticism of the field. It is how theoretical physics develops. But it means there is no first-principles-only gold standard that newcomers must meet to be taken seriously. There is only the question of how few adopted inputs a framework needs before it predicts the rest. The companion papers in this series introduce a framework grounded in the photon baseline as a universal reference state. This first paper asks only that the framework be evaluated by the same standard the field already applies to its existing members.

1. The Standard-of-Evaluation Problem

When a new gravity framework is proposed, by what standard should it be evaluated? The question sounds naive, but it is rarely asked explicitly. The answer is often assumed: a framework should be derived from first principles, contain no free parameters, and predict all observations without adjustment.

Yet no existing gravity framework meets that standard. Three examples make this clear.

General Relativity, as Einstein presented it in 1915, was not derived from deeper principles. The Einstein-Hilbert action was chosen as the simplest scalar Lagrangian consistent with general covariance and second-order field equations. The choice was elegant but it was a choice. The cosmological constant was later added by hand, set to zero, then resurrected when observation required it. Mercury's perihelion advance validated the theory after the form had been chosen. Today the field treats general relativity as foundational. In 1915 it was an ansatz with adopted constants and an adopted functional form.

Modified Newtonian Dynamics, proposed by Milgrom in 1983, was constructed to match galactic rotation curves without invoking dark matter. It introduced one fitted constant, an acceleration

scale of approximately 1.2×10^{-10} meters per second squared, and an interpolating function whose specific form remains a matter of choice within the framework. No derivation of either was offered. The framework predicted the Tully-Fisher relation and the dynamics of ultra-faint dwarf galaxies, predictions that the standard cosmological model has struggled with. The framework's adopted constant does not come from a deeper theory. It comes from fitting to data.

The $f(R)$ family of gravity theories, of which the Starobinsky model is the best known example, is distinguished from general relativity by the choice of a function $f(R)$ replacing the Ricci scalar in the action. The Starobinsky model adopts the specific form $f(R) = R + \alpha R^2$ and fits the constant α to observations of the cosmic microwave background. No claim is made that the functional form is derived from deeper principles. The framework is evaluated by its ability to match cosmological data while avoiding instabilities. The functional form could in principle be anything that satisfies these constraints, and the literature contains many published alternatives.

These three examples are not selected because they are weak frameworks. They are among the most successful and most studied in the field. They are selected because they illustrate a pattern that holds for every gravity framework currently published: each one adopts some structure — a functional form, a symmetry, a coupling constant, a discretization — and derives consequences from the adoption. Each is evaluated by how well its derived consequences match observation.

This is how theoretical physics develops. Forms are proposed, consequences are tested, frameworks survive or die by experimental agreement. The pattern is not a flaw. But its existence means there is no first-principles-only gold standard sitting in the field that new frameworks must meet to be considered seriously. There is only the question of how few adopted inputs a framework requires before it predicts the rest.

This paper surveys twenty-seven published Lagrangian approaches to gravity by what they adopt, what they derive, and what they leave open. It is offered as preparation for evaluating any new framework, including the one introduced in the companion papers of this series, on the same basis the field already uses for its established members.

2. A Survey of Twenty-Seven Gravity Lagrangians by What They Adopt

The frameworks below are grouped by what each adopts as primary input. The grouping is for clarity rather than taxonomy; some frameworks could fit in multiple categories. Each entry states the adoption directly without editorial commentary. The pattern across all groupings is the relevant observation.

2.1 Frameworks That Adopt a Functional Form

Frameworks in this category select a function — of curvature, torsion, or other geometric invariants — and derive field equations from the adopted form. Parameters within the function are constrained by observation.

$f(R)$ gravity (Buchdahl 1970, Starobinsky 1980). Adopts the choice of function $f(R)$. The Starobinsky variant adopts $f(R) = R + \alpha R^2$, with α fitted to observations of cosmic inflation. Why this particular function rather than another is left open within the framework.

Horndeski gravity (1974). Adopts the most general scalar-tensor theory whose field equations

remain second-order. The framework contains four free functions of the scalar field and its kinetic term, which must be chosen or constrained by observation.

Beyond Horndeski / GLPV (Gleyzes, Langlois, Piazza, Vernizzi 2015). Adopts extended scalar-tensor structures with higher-order equations of motion but degenerate higher-order terms to avoid instabilities. The degeneracy condition is adopted, not derived.

DHOST (Degenerate Higher-Order Scalar-Tensor, Langlois 2017). Adopts theories with degenerate Hessian structures to avoid the Ostrogradsky instability. The classification of such theories is mathematically exhaustive within its scope, but the choice to work within this class is itself adopted.

Lovelock gravity (1971). Adopts the most general metric theory with second-order field equations in higher-dimensional spacetimes. In four dimensions, the framework reduces to general relativity plus a topological Gauss-Bonnet term that contributes nothing to the field equations. The justification for higher dimensions is left to be supplied by other considerations.

Gauss-Bonnet extended gravity. Adopts the addition of the Gauss-Bonnet invariant to the Einstein-Hilbert action, with a coefficient that is either fitted to observation or set to zero by hand.

Eddington-Born-Infeld gravity. Adopts a square-root action based on the determinant of the Ricci tensor. The choice of action form is motivated by analogy to Born-Infeld electrodynamics rather than by derivation from deeper principles.

2.2 Frameworks That Adopt Symmetry Structure or Symmetry Breaking

Frameworks in this category choose which symmetries are preserved, broken, or augmented relative to general relativity.

TeVSe (Bekenstein 2004). Adopts a vector field and a scalar field with carefully chosen couplings designed to recover Modified Newtonian Dynamics in the non-relativistic limit while preserving stability in cosmological settings. The vector field carries an adopted unit-norm constraint.

MOG (Moffat 2006). Adopts a massive vector field and a scalar field with coupling constants fitted to galaxy rotation curves and cluster dynamics. The mass of the vector field is fitted, not derived from deeper structure.

Conformal gravity (Mannheim and Kazanas 1989). Adopts Weyl invariance as the primary symmetry principle, with an action constructed from the square of the Weyl tensor. The choice to require Weyl invariance is itself an adoption.

Einstein-Cartan theory. Adopts torsion as an independent geometric field coupled to spin density. In the minimal version, torsion does not propagate because kinetic terms for it are not included. The decision not to include those terms is an adoption.

Cartan-Einstein gravity with quadratic torsion. Adopts quadratic torsion terms in addition to the Einstein-Cartan structure. The coefficients of those quadratic terms are fitted parameters.

2.3 Frameworks That Adopt Coupling Constants and New Sectors

Frameworks in this category introduce new fields or modifications with coupling constants fixed by observation or by stability considerations.

Brans-Dicke (1961). Adopts a scalar field coupled to curvature with a coupling parameter ω . The framework's observational viability requires ω greater than approximately forty thousand, a constraint that produces a fine-tuning question the framework does not address from within.

Scalar-tensor in the Bergmann-Wagoner generalization. Adopts a general coupling function $\omega(\phi)$ and a scalar potential $V(\phi)$. The framework generalizes Brans-Dicke at the cost of leaving the functional forms unconstrained.

Hořava-Lifshitz gravity (2009). Adopts anisotropic scaling between space and time, with the spatial dimension scaling differently from the temporal dimension in the ultraviolet regime. The specific scaling choice is adopted because it makes the theory power-counting renormalizable, not because it follows from deeper principles.

Massive gravity (de Rham, Gabadadze, Tolley 2011). Adopts a specific non-linear potential structure for the graviton mass term, chosen to avoid the Boulware-Deser ghost. The graviton mass parameter is itself fitted.

Bimetric gravity (Hassan and Rosen 2012). Adopts two interacting metrics, one massive and one massless. Generalizes massive gravity at the cost of an additional interaction structure that must be specified.

2.4 Quantum Gravity Approaches

The frameworks below are not classical gravity Lagrangians at the same level as the preceding entries. They are quantization programs or proposals for ultraviolet completion. They share with the classical approaches the feature of adopted structure.

Loop Quantum Gravity (Rovelli, Smolin, Ashtekar). Adopts a background-independent quantization of general relativity using Ashtekar variables. The regularization scheme, including the holonomy variables and area operator, is adopted. The semi-classical limit remains an active research area after more than thirty years.

Causal Dynamical Triangulations (Ambjørn, Jurkiewicz, Loll). Adopts a discretization of spacetime into simplices carrying a causal structure. The Regge action is adopted. The continuum limit remains under study.

Asymptotic Safety (Weinberg, Reuter). Adopts the existence of a non-Gaussian fixed point of the renormalization group flow. The truncation of the infinite-dimensional theory space used to search for this fixed point is itself adopted, and whether the truncation accurately reflects the full theory remains an open question.

String theory low-energy effective action. Adopts a ten-dimensional spacetime, a choice of string vacuum, and a compactification scheme. The low-energy effective action is then derived from these adoptions. The number of viable compactifications is estimated at approximately 10^{500} , none of which is currently selected on principled grounds.

2.5 Teleparallel and Geometric Reformulations

Frameworks in this category reformulate gravity using different geometric primitives than the Riemannian curvature of general relativity.

$f(T)$ teleparallel gravity. Adopts torsion T as the geometric variable, with a chosen function $f(T)$ replacing the simple linear case. Uses the Weitzenböck connection rather than the Levi-Civita

connection.

Symmetric teleparallel $f(Q)$. Adopts non-metricity Q as the geometric variable, with a chosen function $f(Q)$. Uses what is called the coincident gauge.

Palatini $f(R)$. Adopts the metric and connection as independent variables, with the connection then solved for from the field equations. This produces second-order field equations where the metric formulation would give fourth-order equations.

MacDowell-Mansouri gravity (1977). Adopts a gauge-theoretic formulation of gravity with an explicit symmetry-breaking sector. The symmetry-breaking mechanism is chosen rather than derived.

2.6 The Twenty-Seventh Approach

The twenty-seventh framework being introduced in this series is presented in detail in the companion papers. Its adoptions, derivations, and open work are stated as transparently as those of the preceding twenty-six frameworks. It is included in the comparison table in the next section.

3. The Common Structure Across All Twenty-Seven Approaches

The pattern across the twenty-seven frameworks surveyed above is consistent. Each framework adopts some structure as input: a functional form, a symmetry principle, a coupling constant, a discretization, a geometric primitive, or a quantization scheme. From that adopted structure, each framework derives consequences that can be compared to observation. Each framework's standing in the field reflects how well its derived consequences match measurements and how internally consistent its adopted structure is.

This pattern is not a flaw in the field. It is the working method of theoretical physics. Forms are proposed, consequences are tested, frameworks survive or die by experimental agreement. The pattern has produced general relativity, the standard model of particle physics, and the broader edifice of modern fundamental physics. The pattern works.

But the pattern's existence has a methodological implication that is sometimes obscured: there is no framework currently in the field that derives everything from nothing. Every framework adopts something. The standard of evaluation that the field actually uses, in practice, is whether a framework's adopted structure is stated openly, whether its derived consequences agree with observation, and whether its adopted constants are independently measurable. The standard is not zero adoption. The standard is transparent adoption.

This matters for any new framework being considered. A new framework that states its adopted structure openly, derives consequences from it, and submits those consequences to observational test is asking to be evaluated by the same standard the field already applies to its established members. That request is reasonable. It does not require justification beyond pointing out that the standard already exists and is already in use.

4. How Many Measured Numbers Does a Framework Need

The phrase “no free parameters” is used loosely in the literature. A framework may be described as having no free parameters once certain constants are fixed by external measurement. Strictly

speaking, every framework has at least one adopted input, whether a number or a functional form. The honest question is not whether a framework has zero adopted inputs but how few adopted inputs it needs before it predicts everything else.

The table below summarizes adopted inputs for selected frameworks from the survey above, including the twenty-seventh framework introduced in the companion papers of this series.

Framework	Measured puts	In-	Notes
Einstein-Hilbert with Λ	2 (G, Λ)		G fixed by laboratory gravimetry; Λ fixed by cosmological observations of accelerated expansion
Standard Model of particle physics	~ 19 (gauge couplings, Yukawas, CKM, θ_{QCD} , Higgs sector)		All fitted to particle physics experiments; the framework is widely regarded as effective rather than fundamental
Modified Newtonian Dynamics	1 (a_0)		Plus the choice of interpolating function form, which is itself an adoption beyond the single fitted constant
Brans-Dicke	2 (G, ω)		Plus the scalar potential form, which is left unspecified within the minimal framework
$f(R)$ Starobinsky	1 (α)		Plus the choice of functional form $f(R) = R + \alpha R^2$
Loop Quantum Gravity	0 numerical		The regularization scheme and area operator definitions function as adoptions; semi-classical limit is open
String theory effective action	0 beyond string scale	string	The compactification choice from $\sim 10^{500}$ possibilities functions as an adoption
Departure From Photon Baseline	0 numerical		Adopted structure: photon baseline as universal reference state; closure principle; kinematic gap axiom (no configuration at $v = 0$ or $v = c$). Quantitative predictions for specific phenomena are open work in the companion papers.

Strict counting matters less than methodological transparency. Every framework above has some adopted structure. The question for any framework is whether the adoption is named openly, whether the resulting derivations are internally consistent, whether the framework's predictions can be tested against observation, and whether the framework holds itself to the same discipline it asks of others.

5. What Comes Next in This Series

The framework introduced in the companion papers of this series develops the consequences of its adopted structure across the phenomena that gravity theories address. Each phenomenon receives its own focused paper. The annotated tour below identifies what each paper addresses and what structural claim it develops. The previews are structural descriptions, not derivations. Readers seeking specific topics can navigate directly to the relevant papers; readers seeking the full picture can follow the series in order.

5.1 Tier 1 — Foundational Papers

Paper 2 develops the framework’s geometric foundation. The three-axis interaction fabric, the twenty-seven possible configurations arising from three discrete values per axis, and the framework’s primary geometric primitive are established. The paper sets the language used throughout the rest of the series.

Paper 3 presents the nine-gluon classification by inside-outside content structure. The classification organizes the twenty-seven configurations into nine categories distinguished by how their content distributes between the central vertex and the outer regions. The closed circuit at the center defines the photon state; departures from that closure define every other configuration.

Paper 4 establishes the photon as the baseline reference state and explains why the electron-positron pair represents the first stable departure from baseline through axis tilt. The 720-degree rotation requirement for fermions is derived geometrically from the tilt history rather than postulated as a quantum property.

Paper 5 presents the closure principle that governs configurational stability. Configurations close their geometric loops or they decay until they do. The principle determines which states persist, which transform, and which dissolve into their constituents.

Paper 6 establishes rest, uniform motion, and inertia as active internal ledger states rather than passive properties. Newton’s First Law receives a mechanistic underpinning: locked-in directional tilt parameters in the configuration’s internal accounting maintain motion without external forcing.

Paper 7 develops acceleration, momentum, and geometric consolidation as the integration of forcing geometry into a configuration’s internal ledger. The plasma rocket example illustrates how added geometry shifts the configuration toward the photon baseline and how consolidation back to baseline produces deceleration.

Paper 8 establishes mass as the internal accounting of a configuration’s departure from the photon baseline speed. Mass is not a substance; it is the geometric record of how far a configuration sits from its reference state.

Paper 9 establishes heat as the interaction event itself rather than as a substance or field. Heat occurs at the point of contact between configurations; it does not propagate as an independent quantity.

Paper 10 establishes gravity as the fabric’s recording of departure from the photon baseline and as the resolution-seeking activity of configurations searching for consolidation paths. The same mechanism that drives cosmic expansion at large scales produces apparent gravitational attraction at smaller scales where consolidation paths are accessible.

Paper 11 establishes the distinction between simple states and complex configurations and identifies which side of any interaction performs the active resolution-seeking work. Simple states are integrated by complex configurations searching for consolidation paths, not the other way around.

Paper 12 establishes the neutrino as bankrupt geometric residue from high-energy decay events. The configuration locks into a near-baseline state with insufficient transactional energy to consolidate further, providing the framework's fourth manifestation of resolution-seeking alongside expansion, attraction, and decay.

Paper 13 establishes time as the ledger of 720-degree rotation pattern progress in tilted configurations, with photon configurations having no rotation ledger and therefore no time registering for them. The existence of bankrupt configurations provides a framework-native account of the arrow of time and the non-linearity of the universe.

Paper 14 develops resolution-seeking as the mechanism of cosmic expansion. Configurations at large scales seeking consolidation paths that are not available contribute their resolution-seeking activity to the fabric as the observable expansion of space.

Paper 15 establishes beta decay as a continuous mechanism converting one tilt direction into another, dissolving the matter-antimatter asymmetry into a steady-state result of ongoing geometric processes and providing the framework's account of where neutrinos originate.

Paper 16 establishes mesons, baryons, and tetraquarks as specific configurational types distinguished by their geometric closure structure. The standard model's flavor and color classifications are reframed as observer-side labeling of underlying geometric configurations.

Paper 17 reinterprets the Higgs phenomenon as a ledger reconciliation event rather than as a mass-giving particle interaction. The 125 GeV observation receives a framework-native account that retains historical continuity with the standard interpretation while operating at a different mechanistic level.

5.2 Tier 2 — Critique Papers

Paper 18 examines the methodology of particle collider inference, distinguishing what is directly measured in collision experiments from what is inferred through theoretical frameworks. The framework's discipline of separating observation from inference is applied to high-energy physics experimental practice.

Paper 19 examines the double-slit experiment and identifies the unspecified variable — the photon's geometric spatial extent — whose unspecification underwrites the standard superposition interpretation. The framework's alternative interpretation does not require superposition as a fundamental concept.

Paper 20 examines the cosmic microwave background measurement chain from photon-detector interaction events through the layered inferences that produce statements about early-universe temperature. The framework's observation-versus-inference discipline is applied to one of cosmology's most cited measurements.

Paper 21 examines magnetars as a case study in naming-versus-observation. What is directly observed is photon emission with specific spectral characteristics; the magnetic field interpretation is inferred under standard electromagnetic theory. The framework's primitives account for the observations without requiring the magnetic interpretation as fundamental.

Paper 22 examines the LHCb tetraquark observations and develops the framework’s structural account of how added geometry from proton-proton collisions combines to produce transient tetraquark configurations that decay through the framework’s closure mechanism.

Paper 23 examines gravitational lensing observations and develops the framework’s structural account of why mass produces geometric effects on light propagation. The framework’s resolution-seeking mechanism provides the underlying cause that general relativity describes geometrically without explaining.

Paper 24 examines gravitational wave observations and develops the framework’s merger audit account. Detected waves are read as the outgoing radiation from geometric reconciliation events at merger, with the post-merger remnant carrying the unresolved geometry that could not return to baseline.

Paper 25 examines neutrino oscillation observations and applies the framework’s methodology to distinguish the static configuration in transit from the distance-dependent detector audits that produce the appearance of flavor change at detection.

Paper 26 examines cosmological redshift measurements and identifies the inference chain from electron signatures in CCD sensors through layered theoretical assumptions about photon propagation and cosmological expansion. The framework offers a candidate alternative interpretation operating at a different theoretical level.

5.3 Tier 3 — Bridge and Capstone

Paper 0 documents the methodological journey from the earlier research project, in which a fitted parameter was extended across 137 papers without ever being derived, to the current framework’s discipline of grounded derivation from a minimal adopted structure. The autobiographical bridge serves as documented evidence of the methodological discovery the series develops.

Compendium consolidates the framework’s content into a single citable reference document for readers seeking the unified picture without working through each paper in the series. The compendium is written after the framework’s papers are established and serves as the canonical entry point for citation.

The series develops a coherent program from a small set of adopted inputs. Each paper makes a focused contribution that can be evaluated on its own merits and cross-references the others through the citation network. The framework does not claim to have derived every observed phenomenon quantitatively; it claims structural consistency from a smaller adoption set than the surveyed frameworks above, with quantitative development as research direction rather than completed work. The framework asks to be evaluated by the standard the field already applies to its members: transparent adoption, derived consequences, observational testability, and methodological discipline.

If the framework’s reduced set of adopted inputs produces consequences that agree with observation across the phenomena it addresses, the framework earns the consideration that any framework in the survey above receives. If it does not, the framework is falsified by the same standard. This paper does not argue that the framework is correct. It argues that the framework is asking to be evaluated by the rules the field already uses.

6. Summary

Every gravity framework currently in the field adopts some structure as input — a functional form, a symmetry, a coupling constant, a discretization, a geometric primitive. From the adopted structure, each framework derives consequences that are tested against observation. The pattern is consistent across all twenty-seven approaches surveyed in this paper. There is no first-principles-only gold standard currently in use that excludes new frameworks on principle.

The standard the field actually uses, in practice, is whether a framework’s adopted structure is stated openly, whether its derived consequences agree with observation, and whether its adopted constants are independently measurable. The twenty-seventh framework being introduced in this series adopts a smaller set of inputs than any of the surveyed frameworks: zero numerical constants, one reference state (the photon baseline), one geometric principle (closure), and one axiom (the kinematic gap). The framework’s consequences are developed in the companion papers and submitted to observational test by the same standard the field already applies.

The next paper in this series introduces the framework’s geometric structure and the closure principle that organizes its configurations.

Departure From the Photon Baseline: A Universal Ruler for Mass, Gravity, and Time
 Paper 1 of the series. Author: Dervis Kadunic. Draft version for review.